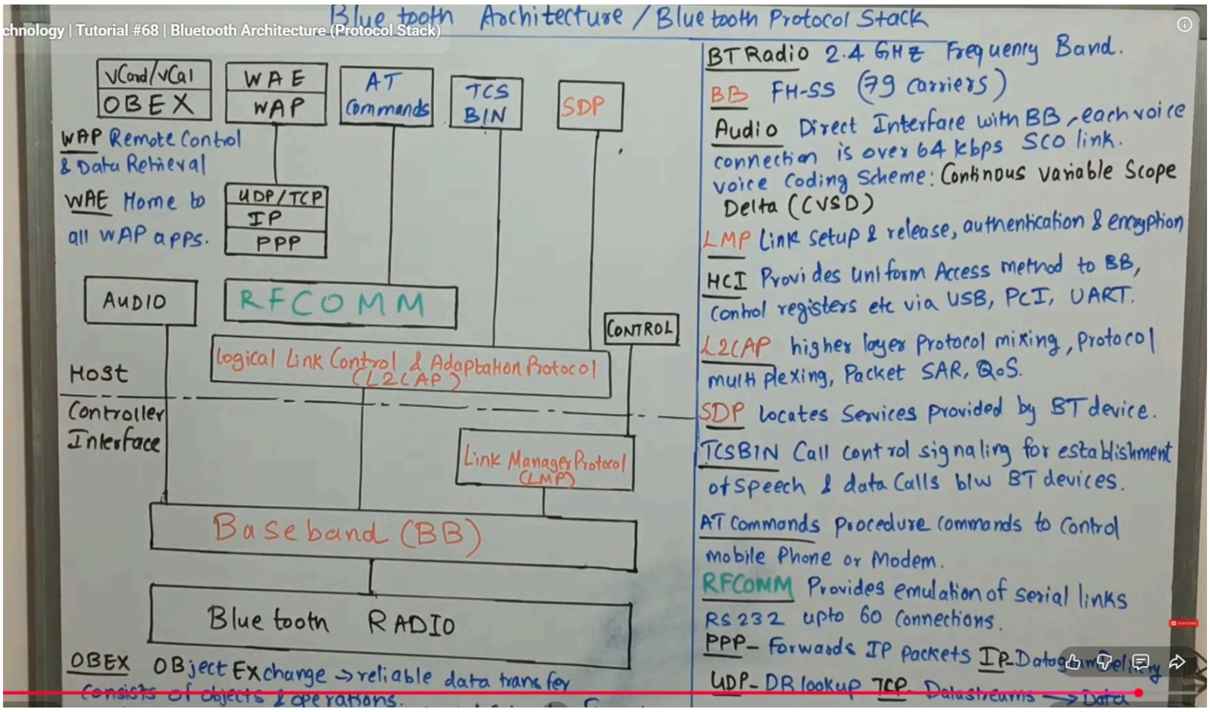

📡 Bluetooth Architecture / Protocol Stack

Introduction

Bluetooth architecture is divided into **layers** that manage communication between devices. It consists of **Controller part** and **Host part**, connected through **HCI (Host Controller Interface)**.

🔸 1. Bluetooth Radio Layer

- Operates in **2.4 GHz ISM band**
- Uses **Frequency Hopping Spread Spectrum (FHSS)**
- 79 channels used
- Responsible for **wireless transmission**

🔸 2. Baseband (BB)

- Handles **physical channel and link control**
- Supports:
  - **SCO (voice link)**
  - **ACL (data link)**
- Performs **error correction and packet formation**

🔸 3. Link Manager Protocol (LMP)

- Responsible for:
  - Link setup and release
  - Authentication
  - Encryption
  - Power control

🔸 4. Host Controller Interface (HCI)

- Interface between **host and controller**
- Provides uniform access via:
  - USB
  - UART
  - PCI

🔸 5. L2CAP (Logical Link Control and Adaptation Protocol)

- Multiplexes higher layer protocols
- Provides:
  - Segmentation and reassembly (SAR)
  - Quality of Service (QoS)

🔶 6. RFCOMM

- Emulates **serial port communication (RS-232)**
- Supports multiple connections

🔶 7. SDP (Service Discovery Protocol)

- Used to **discover services** offered by Bluetooth devices

🔶 8. TCS (Telephony Control Protocol)

- Used for **call control signalling**
- Establishes voice/data calls

🔶 9. AT Commands

- Used to control:
  - Mobile phone
  - Modem operations

🔶 10. OBEX (Object Exchange Protocol)

- Used for **file transfer**
- Ensures reliable data exchange

🔶 11. PPP, TCP/IP, UDP

- Enable **internet communication over Bluetooth**
- PPP → forwards IP packets
- TCP/UDP → transport protocols

🔸 12. Audio Layer

- Directly interfaces with baseband
- Used for **voice transmission**

🧠 Quick Summary (Exam Tip)

- **Radio + Baseband → Physical communication**
- **LMP → Link control & security**
- **HCI → Interface**
- **L2CAP → Data handling**
- **RFCOMM / SDP / OBEX → Applications**